SCOPE-Fin: Budget-Aware Orchestration of Retrieval, Quantitative Reasoning, and Risk Triage for Financial Research

A Selective-Routing Framework and Evaluation Protocol

Xiang Zhang
College of Computer Science, Sichuan University
2024324022023@stu.scu.edu.cn

October 2026

Abstract

Financial research assistants face heterogeneous requests: some need a single fact lookup, others require multi-document synthesis, numerical calculation, or temporal transaction analysis. Applying the most expensive model to every request wastes computation, while routing every request through a lightweight model risks inaccurate or unsupported answers. This paper proposes SCOPE-Fin (Selective Cost-aware Orchestration for Processing Evidence in Finance), a framework that selects specialist reasoning routes according to task type, expected information gain, evidence quality, and a per-request budget. Its route library combines direct retrieval, multi-hop financial RAG, symbolic numerical execution, and temporal risk scoring; a quality-cost gate either accepts a supported draft, invokes another route, or escalates the case for analyst review. The design synthesizes ideas from three supplied papers on blockchain anomaly detection, financial document QA, and modular mathematical reasoning, but shifts the research question from evidence integration to selective allocation of computation and human attention. We formalize the routing objective, define a benchmark-based evaluation protocol, and state testable hypotheses concerning answer quality, cost, and review burden. A heatmap summarizes the ablation values reported in the supplied FinLLaMA-RAG preprint as contextual evidence only. This manuscript reports no new SCOPE-Fin experiments; source-reported values are not presented as results for the proposed framework. The intended use is analyst decision support, with explicit escalation for uncertain or high-impact cases.

Keywords: financial research; adaptive routing; computational budget; retrieval-augmented generation; symbolic reasoning; human-AI collaboration; selective prediction

1 Introduction

Financial research systems are asked to perform tasks with very different computational demands. Looking up a dated disclosure may require only targeted retrieval, while reconciling values across a report may need multiple retrieval hops and executable arithmetic. A transaction-risk question can instead require temporal modeling over a sequence of events. Treating all of these requests as one generic text-generation problem can spend computation on easy cases and under-serve difficult ones.

Retrieval-augmented generation (RAG) provides a way to condition model output on external documents [5]. Financial QA benchmarks such as FinQA and TAT-QA illustrate that answering questions over reports can involve both numerical operations and mixed text-table evidence [6, 7]. Meanwhile, blockchain transaction analysis involves different inputs and objectives: methods such as the Elliptic graph-learning work and the uploaded GAM-CoT paper focus on identifying suspicious temporal or relational patterns [4, 1]. A single pipeline is unlikely to be the most economical route for every task.

This paper proposes SCOPE-Fin, a selective orchestration framework for a financial research assistant. It uses a task profiler to estimate the request type and difficulty, a policy to choose a subset of specialist routes, and a quality-cost gate to decide whether to stop, spend more computation, or request human review. The route library draws on the three supplied papers: temporal risk scoring from GAM-CoT [1], multi-hop retrieval and context fusion from the FinLLaMA-RAG preprint [2], and modular symbolic-neural problem decomposition from MetaMath-LLaMA [3]. In contrast to a fixed end-to-end workflow, SCOPE-Fin makes resource allocation and escalation policy the central design questions.

The proposed work makes four contributions:

1. It formulates financial assistant routing as a constrained decision problem over answer quality, computation cost, and human-review effort.
2. It defines four specialist routes and a quality-cost gate that can add analysis selectively instead of invoking every tool on every request.
3. It specifies an evaluation protocol that measures quality at matched compute budgets, including selective accuracy and analyst review burden.
4. It separates the proposal from prior reported data: the included ablation visualization transcribes a supplied paper's table and does not claim a SCOPE-Fin result.

2 Background and Motivation

2.1 Specialist components in financial AI

The GAM-CoT paper describes feature- and time-aware attention for transaction anomaly detection. Its target is a structured event sequence, not a general document question [1]. The FinLLaMA-RAG preprint instead describes dense retrieval, iterative multi-hop reasoning, and context aggregation for document-level financial QA [2]. MetaMath-LLaMA proposes a modular scheduler and a symbolic-neural computation path for interpretable multi-step mathematics [3]. These papers motivate complementary specialists, but their task settings do not by themselves define when a deployed assistant should pay for each route.

The evaluation datasets also remain task-specific. Elliptic-style transaction labels measure transaction risk classification; FinQA and TAT-QA measure question answering over financial reports. They should not be collapsed into one score because they assess different outputs and error costs [4, 6, 7]. SCOPE-Fin therefore treats route selection as a separate, measurable layer above the task-specific models.

2.2 Selective prediction and human review

Selective prediction studies systems that may withhold an answer when confidence is insufficient, trading answer coverage against risk [11]. Human-AI interaction guidance emphasizes keeping people informed about system limitations and providing a way to correct or override automated behavior [10]. SCOPE-Fin applies these ideas operationally: an answer is not accepted solely because a language model is confident; route evidence, calculation checks, and task-specific validation contribute to the stopping decision. Reasoning-and-action systems such as ReAct and chain-of-thought prompting motivate decomposition and tool use, while SCOPE-Fin focuses on whether and when those steps are worth their cost [9, 8].

2.3 Source-reported ablation evidence

The supplied FinLLaMA-RAG preprint reports a full-model ablation table with four metrics. Its table lists nDCG@10 of 0.62, BLEU of 30.5, ROUGE-L of 35.2, and F1 of 0.75 for the full system; its retrieval-only and generation-only rows report lower scores on each listed metric. Figure 1 shows the raw values and each ablation's ratio to the full-system score. This pattern is consistent with the value of combining retrieval and generation in that reported setup, but it does not establish that every request needs the full pipeline, nor does it establish SCOPE-Fin's effectiveness. The values are transcribed from the uploaded preprint and have not been independently reproduced; the paper does not provide uncertainty estimates alongside this table.

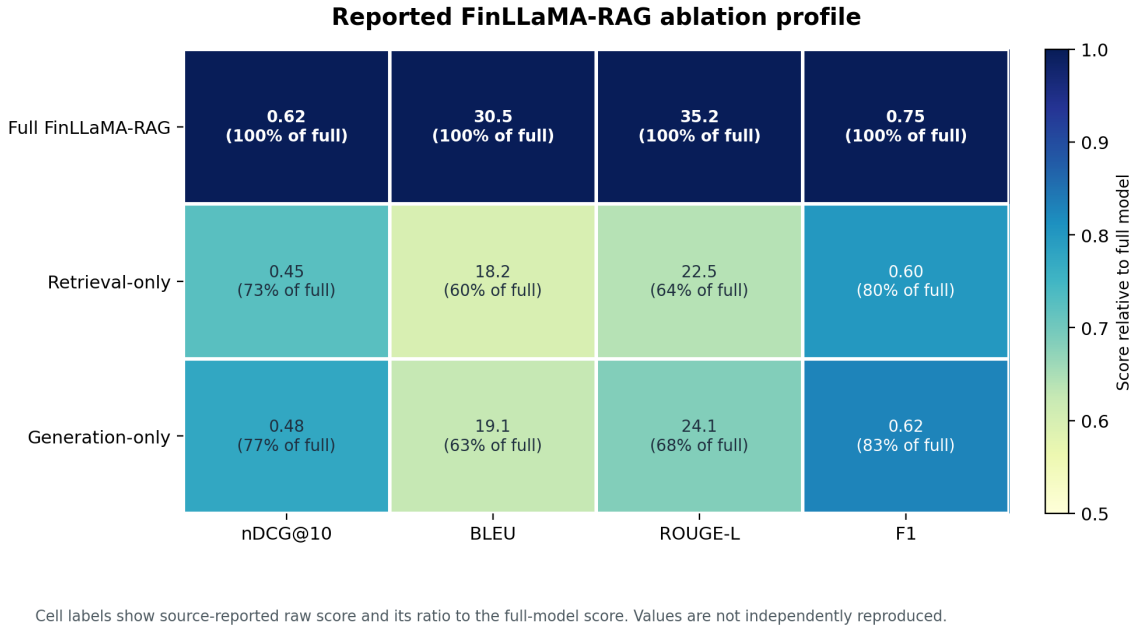


Figure 1: Ablation scores reported in the supplied FinLLaMA-RAG preprint. Cells show raw values and ratios relative to the full model; ratios are calculated from the reported table. They are contextual source data, not SCOPE-Fin results.

3 Problem Definition

Let a request be q, available evidence be x, and remaining computation budget be B. SCOPE-Fin has a finite route set $\mathcal{R} = \{r_0, r_1, r_2, r_3\}$: direct evidence lookup, multi-hop document QA, symbolic quantitative reasoning, and temporal transaction-risk analysis. A request may invoke one route or a compatible sequence of routes. After each route, the system observes an intermediate state z_t containing retrieved evidence, operation results, estimated support, and spent cost.

A policy π maps the request and current state to the next action: invoke a route, return a draft, or escalate to an analyst. We define the objective as

$$\pi^* = \arg\max_{\pi} \ \mathbb{E}_{q,x}[U(y, q, x) - \lambda_c C(\pi) - \lambda_h H(\pi)] \quad \text{subject to} \quad C(\pi) \le B, \ R_{\text{unsupported}}(\pi) \le \delta, \tag{1}$$

where U is task utility, C is computational cost, H is human-review effort, and $R_{\text{unsupported}}$ is the rate of unsupported claims among accepted answers. The weights λ_c and λ_h encode deployment priorities, while δ is a pre-specified safety tolerance. These quantities must be reported separately in evaluation; a single aggregate score can conceal a costly or poorly grounded policy.

4 SCOPE-Fin Architecture

4.1 Task profiler and route library

The task profiler extracts a compact feature vector $\psi(q, x)$: requested output type, presence of numeric operators, number of apparent source entities, temporal span, retrieval ambiguity, and impact category. It assigns a provisional route distribution, not a final answer. Each specialist returns a structured result with its inputs, output, cost, and route-specific diagnostics.

Table 1: SCOPE-Fin route library and intended activation conditions.

Route	**Typical trigger**	**Output**	**Cost profile**
Direct lookup	One fact, known source, low ambiguity	Source-linked fact or a request for clarification	Low
Multi-hop document QA	Cross-section or cross-document synthesis	Answer with retrieved passages and hop trace	Medium to high
Symbolic quantitative reasoning	Arithmetic, comparison, unit conversion, or constraint check	Executable expression and validated value	Low to medium
Temporal risk analysis	Ordered transaction or event sequence with risk labels or indicators	Risk score and salient temporal window	Medium to high
Analyst escalation	Conflicting sources, insufficient support, or high-impact uncertainty	Human decision, correction, or additional evidence request	Human-time dependent

Figure 2 depicts the workflow. Route outputs are evaluated by a gate that checks support, task-appropriate validation, remaining budget, and whether an additional route is likely to change the decision. The final research memo separates sourced facts, calculated values, model-derived risk signals, and unresolved questions.

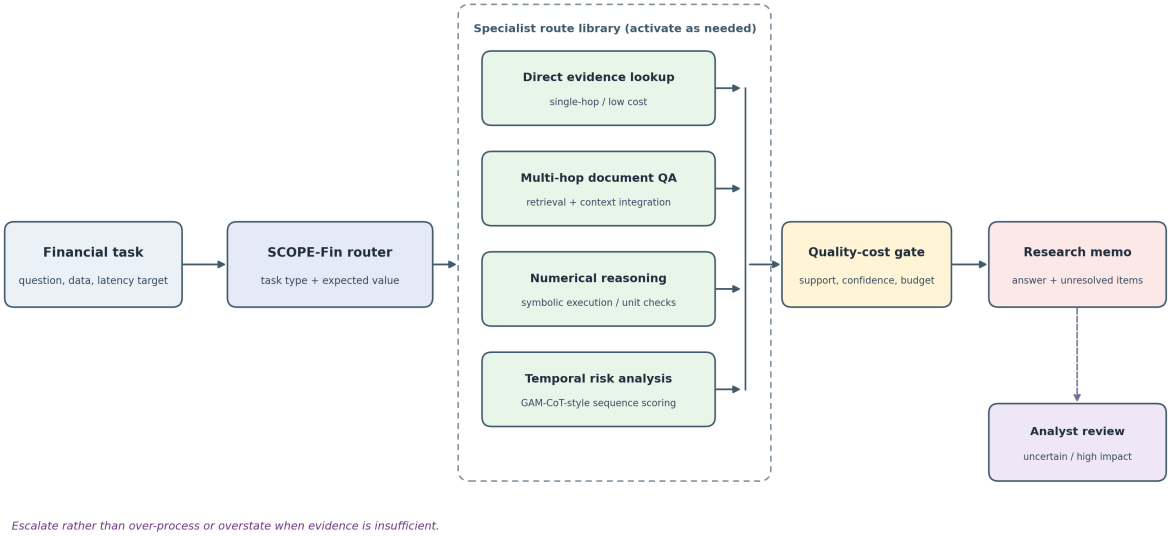


Figure 2: SCOPE-Fin selectively activates specialist routes according to task type and expected value of additional analysis. Low-confidence or high-impact cases are directed to analyst review.

4.2 Value-of-computation routing

For candidate route r at state z_t, let $\widehat{\Delta U}(r \mid q, z_t)$ estimate the expected improvement in task utility, $\widehat{\Delta C}(r)$ estimate incremental computational cost, and $\widehat{\Delta H}(r)$ estimate the effect on review effort. A budget-aware decision score is

$$V(r \mid q, z_t) = \widehat{\Delta U}(r \mid q, z_t) - \lambda_c \widehat{\Delta C}(r) - \lambda_h \widehat{\Delta H}(r). \tag{2}$$

The policy selects a compatible route with positive value, subject to the remaining budget. It stops when no available route has positive estimated value, when the evidence-support condition is met, or when the budget is exhausted. Cost estimates may include model tokens, retrieval calls, accelerator time, wall-clock latency, and specialist-tool execution. The deployment must state which cost measure is optimized; token count alone may not reflect total operating cost.

The route policy is trained or calibrated on development data only. A simple initial implementation can use task-type rules and validation-set cost/quality curves; a learned policy is justified only if enough labeled routing examples exist. Oracle routing, derived from gold task labels, is retained as an upper-bound comparison, not as a deployable baseline.

4.3 Acceptance, continuation, and escalation

The gate maintains separate signals: evidence coverage, source agreement, numeric execution validity, route confidence, and impact level. A numerical answer requires a reproducible expression and unit check. A document claim requires cited source spans. A temporal score remains explicitly marked as model-derived and is not converted into a factual allegation. Confidence thresholds are calibrated by task family on held-out validation data.

The system continues analysis when the expected benefit in Equation 2 exceeds its cost and budget remains. It escalates when sources conflict, required evidence is absent, a high-impact request has low calibrated confidence, or independent routes disagree materially. Human feedback can correct labels, clarify intent, or identify missing documents; it should not be silently folded into future training without governance and consent.

5 Evaluation Protocol

5.1 Tasks and datasets

The evaluation separates route quality from route-selection quality. Elliptic is used for temporal transaction-risk tasks, while FinQA and TAT-QA are used for financial numerical and text-table QA [4, 6, 7]. A new mixed-request test set is needed to evaluate routing decisions across task types. It should contain queries labeled with acceptable routes, source requirements, difficulty, impact, and review need. Those labels must be independently annotated and adjudicated; this paper does not claim such a dataset has already been collected.

Table 2: Proposed test matrix and primary measurements.

Test set	**Capability**	**Primary measures**	**Leakage / validity control**
Elliptic [4]	Temporal transaction scoring	AUPRC, minority recall at fixed FPR, calibration, latency	Preserve temporal order and avoid future-to-past information.
FinQA [6]	Numerical report QA	Exact match, execution accuracy, symbolic error rate	Separate reports between splits; normalize units and rounding.
TAT-QA [7]	Text-table QA	Answer F1, evidence-cell recall, tool-call rate	Score table and text evidence independently.
Mixed routing set	Route selection and escalation	Quality at fixed cost, cost at fixed quality, route accuracy, review time	Hold out entities, source reports, and query templates.

5.2 Comparators and ablations

Compare five policies: (1) a low-cost single-route baseline; (2) an always-on pipeline that invokes every relevant specialist; (3) a fixed rule-based router; (4) a learned SCOPE-Fin policy; and (5) an oracle policy with gold route labels. Also compare automatic-only operation with selective analyst escalation. Ablations remove expected-value estimation, cost estimation, task-specific confidence calibration, or the escalation action. The same language model, corpus, retrieval index, and answer format should be held constant where possible.

5.3 Metrics and statistical analysis

Report answer quality separately by task: execution accuracy and exact match for numerical QA, evidence-supported answer F1 for document QA, and AUPRC plus minority-class recall for transaction risk. For the router, report the quality-cost frontier, cost at pre-defined non-inferior quality levels, route-selection accuracy, escalation coverage, selective risk, end-to-end latency, and human correction time. Measure cost with at least two operational units, such as model-token usage and wall-clock latency, and disclose the hardware/API conditions.

Use entity- and time-disjoint splits when appropriate. Select thresholds and cost weights on validation data, then freeze them before test evaluation. Compare paired systems with bootstrap confidence intervals over requests and report per-task outcomes, not only pooled averages. Human reviewers should be blinded to policy identity when rating answer support and edit effort. Determine sample size for any analyst study using a pilot-based power analysis; do not infer user-study effects from benchmark metrics alone.

5.4 Research questions and hypotheses

RQ1: Can SCOPE-Fin reduce computation at matched answer quality compared with an always-on specialist pipeline? **RQ2:** Which observable request features best predict when an additional retrieval or symbolic step changes the answer? **RQ3:** Does selective escalation improve supported-answer reliability without transferring excessive workload to analysts?

We propose three preregistered hypotheses: **H1**, at a pre-specified non-inferiority margin, selective routing uses less computation than the always-on policy while preserving task quality; **H2**, routing numerical questions to symbolic execution reduces arithmetic error versus direct generation; and **H3**, escalation improves selective accuracy at matched answer coverage on mixed-request tasks. These remain hypotheses until the specified comparisons are conducted.

6 Discussion and Limitations

SCOPE-Fin treats computation as an allocatable resource rather than an unlimited background service. The source-reported ablation profile in Figure 1 suggests that combining retrieval and generation can be useful in the uploaded study, while the route policy asks whether that combination is necessary for a given request. This distinction motivates a budget-quality evaluation rather than a single full-system score. The approach may be most useful where requests vary substantially in difficulty and where a human analyst can review the uncertain tail.

The framework has important limitations. Route-value estimates can be miscalibrated, and a policy may systematically under-allocate effort to unfamiliar or minority cases. Benchmarks do not capture every reporting convention, jurisdiction, or analyst workflow. The uploaded source studies also have different datasets and experimental designs; their reported scores cannot be compared as if they came from a shared test set. Human escalation can reduce automation risk but may add delay and operational burden. Finally, model-derived risk rankings are not findings of intent, fraud, or criminality. Any deployment requires privacy controls, auditability, local validation, and human accountability.

7 Conclusion

This paper proposed SCOPE-Fin, a budget-aware routing framework for financial research tasks. It combines direct lookup, multi-hop financial RAG, symbolic quantitative execution, and temporal risk analysis, then decides whether to stop, invoke another route, or escalate to an analyst. Its key research claim is testable: selective orchestration should be evaluated on the quality-cost frontier and on human review burden, rather than presumed effective because specialist components perform well in isolation. The supplied papers motivate the route library; the proposed benchmarks and hypotheses define the next empirical steps. No SCOPE-Fin performance claim is made without those experiments.

References

[1] X. Huang, C. Zhao, X. Li, C. Feng, and W. Zhang, "GAM-CoT Transformer: Hierarchical Attention Networks for Anomaly Detection in Blockchain Transactions," *Journal of Emerging Applied Artificial Intelligence*, 2025.

[2] X. Huang, Z. Lin, F. Sun, W. Zhang, K. Tong, and Y. Liu, "Enhancing Document-Level Question Answering via Multi-Hop Retrieval-Augmented Generation with LLaMA 3," TechRxiv preprint, posted June 25, 2025, doi: 10.36227/techrxiv.175086164.45839186/v1.

[3] X. Huang, Z. Wang, X. Liu, Y. Tian, and Q. Leng, "Towards Interpretable and Consistent Multi-Step Mathematical Reasoning in Large Language Models," manuscript provided by the authors, 2025.

[4] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson, "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics," arXiv:1908.02591, 2019.

[5] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[6] Z. Chen *et al.*, "FinQA: A Dataset of Numerical Reasoning over Financial Data," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 3697–3711, 2021, doi: 10.18653/v1/2021.emnlp-main.300.

[7] F. Zhu, W. Lei, Y. Huang, C. Wang, S. Zhang, J. Lv, F. Feng, and T.-S. Chua, "TAT-QA: A Question Answering Benchmark on a Hybrid of Tabular and Textual Content in Finance," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pp. 3277–3287, 2021, doi: 10.18653/v1/2021.acl-long.254.

[8] J. Wei *et al.*, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[9] S. Yao *et al.*, "ReAct: Synergizing Reasoning and Acting in Language Models," in *International Conference on Learning Representations*, 2023.

[10] S. Amershi *et al.*, "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, paper 3, pp. 1–13, 2019, doi: 10.1145/3290605.3300233.

[11] Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.